

SUSMA R

DATA SCIENTIST / ML ENGINEER / AI ENGINEER

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SUMMARY

Proficient in Machine Learning, Data Analysis, and AI, with expertise in Python, SQL, and TensorFlow and PyTorch. Experienced in developing predictive models and passionate about leveraging data to solve real-world problems. Adept at applying analytical skills to drive insights and inform decision-making processes. Passionate about transforming data into insights to tackle real-world problems.

WORK EXPERIENCE

Data Science Internship, Bosch Global Software Technologies (Bangalore) Jan 2024 - May 2024

Worked on Responsible AI to create End-to-End Robustness evaluation for AI/ML models. Utilized advanced evaluation techniques including Adversarial testing, Randomized smoothing, Counterfactual Explanation, and Out-of-distribution detection. Employed FastAPI for API creation, ensuring seamless integration with the frontend for model evaluation and visualization.

Tech Stack : Python, FastAPI

Project Trainee, Bosch Global Software Technologies (Coimbatore) Jul 2022 - Nov 2022

Automated data collection from JIRA server for live ticket status and metrics extraction. Implemented interactive RNM (Metrics) dashboard, displaying real-time updates on ticket priorities, productivity, and quality. Engineered actionable insights into ticket management, empowering employees with informed decision-making capabilities.

Tech Stack : Python, JIRA, Plotly, Dash

PROJECTS

Legal Information Retrieval System

Developed an AI-powered system using Python and OpenAI to translate user queries into relevant legal information, integrating Llama Index, Chainlit, and a local vector database. Implemented tools for extracting key legal concepts and identifying corresponding clauses in updated legal texts, ensuring precision and relevance.

Tech Stack : Python, OpenAI, Llama Index, Chainlit, Local Vector Database

Real-Time Squat Tracking and Analysis System

This project implements a real-time squat tracking system with under 100ms latency using Flask, OpenCV, and MediaPipe's 33 keypoint pose estimation. It processes 1080p video, calculates joint angles (hip, knee, ankle) with 95% accuracy, and tracks squats at 30 FPS. Keypoints and angles are displayed live, enabling advanced movement analysis for fitness applications.

Tech Stack : Python, HTML, Flask (Python), OpenCV, MediaPipe

Weather Anomaly Detection System

Engineered a weather anomaly detection system leveraging Local Outlier Factor (LOF) with novelty detection. Conducted comprehensive data preprocessing and scaling using Python. Developed a Streamlit application for interactive anomaly detection and integrated Explainable AI (XAI) using LIME for model interpretation and feature importance analysis.

Tech Stack : Python, Pandas, NumPy, Scikit-learn, Streamlit, LIME, VS Code

SKILLS

Areas of Interest	Machine Learning, Deep Learning, NLP, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Statistical Analysis, Web Technology, Database Management System
Programming Languages	Python, SQL, C
Frameworks and Platforms	Tensorflow, Pytorch, Scikit-learn, Langchain, OpenCV, Flask, FastAPI
Database	MySql, MongoDB, VectorDB
Tools	PowerBI, Tableau
Web Development	HTML, CSS, JavaScript, React.js, Node.js

EDUCATION

M.Sc. Decision and Computing Sciences(Integrated), Coimbatore Institute of Technology	2024
(CGPA: 8.69)	
Grade - 12, Sri Ramakrishna Matriculation Higher Secondary School	2019
(Percentage: 79.5%)	
Grade - 10, Sri Ramakrishna Matriculation Higher Secondary School	2017
(Percentage: 98%)	

PUBLICATIONS

“A Real Time Web-Based Flask Application For Detection Of Facial Emotions Using Deep Learning Model” (Apr. 2023). In: *JASC*.

“Mushroom Classification To Detect Their Edibility using ML Algorithms and Rapid Miner” (Apr. 2023). In: *IRJMETs*.

“Comprehensive Framework for Robustness Evaluation on Numeric Data Classification” (June 2024). In: *ICCCNT 2024 held at IIT-Mandi, IEEE Electronics Packaging Society*.

CERTIFICATES

- Data Analytics with Python - NPTEL
- Tools for Data Science - Coursera
- SQL (Advanced) - HackerRank